

LPPool3d#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.pooling.LPPool3d>.

Description:

Applies a 3D power-average pooling over an input signal composed of several input planes. On each window, the function computed is: At $p = \infty$, one gets Max Pooling At $p = 1$, one gets Sum Pooling (which is proportional to average pooling) The parameters `kernel_size`, `stride` can either be:

Function Signature

```
class torch.nn.modules.pooling.LPPool3d(norm_type,  
kernel_size, stride=None, ceil_mode=False)[source]#
```

Parameters

Shape:

```
forward(input)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> # power-2 pool of square window of size=3, stride=2
>>> m = nn.LPPool3d(2, 3, stride=2)
>>> # pool of non-square window of power 1.2
>>> m = nn.LPPool3d(1.2, (3, 2, 2), stride=(2, 1, 2))
>>> input = torch.randn(20, 16, 50, 44, 31)
>>> output = m(input)
```

Notes & Warnings

NOTE If the sum to the power of p is zero, the gradient of this function is not defined. This implementation will set the gradient to zero in this case.

Detailed Documentation

Documentation

Applies a 3D power-average pooling over an input signal composed of several input planes.

On each window, the function computed is:

At $p = \infty$, one gets Max Pooling

At $p = 1$, one gets Sum Pooling (which is proportional to average pooling)

The parameters `kernel_size`, `stride` can either be:

a single int – in which case the same value is used for the height, width and depth dimension

a tuple of three ints – in which case, the first int is used for the depth dimension, the second int for the height dimension and the third int for the width dimension

If the sum to the power of p is zero, the gradient of this function is not defined. This implementation will set the gradient to zero in this case.

`kernel_size` (Union[int, tuple[int, int, int]]) – the size of the window

`stride` (Union[int, tuple[int, int, int]]) – the stride of the window. Default value is `kernel_size`

`ceil_mode` (bool) – when True, will use ceil instead of floor to compute the output shape

Input: (N,C,D_{in},H_{in},W_{in})(N, C, D_{in}, H_{in}, W_{in})(N,C,D_{in},H_{in},W_{in}) or (C,D_{in},H_{in},W_{in})(C, D_{in}, H_{in}, W_{in})(C,D_{in},H_{in},W_{in}).

Output: (N,C,D_{out},H_{out},W_{out})(N, C, D_{out}, H_{out}, W_{out})(N,C,D_{out},H_{out},W_{out}) or (C,D_{out},H_{out},W_{out})(C, D_{out}, H_{out}, W_{out})(C,D_{out},H_{out},W_{out}), where

Runs the forward pass.
